Prof. Ashmeet Singh
ashmeet@iitd.ac.in
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Standing Waves, Boundary Conditions, and Fourier Series

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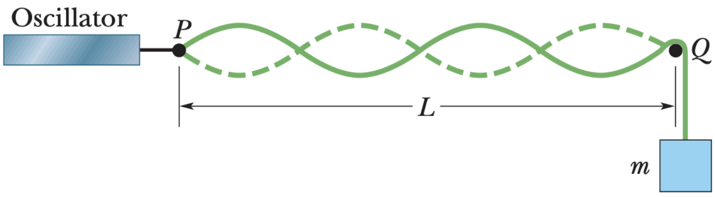
1 *Standing Waves! [Basic]

In the following figure, a string tied to a sinusoidal oscillator (similar to what I showed you in class) at P and running over a support at Q, is stretched by a block of mass m. Separation between P and Q is $L = 1.2$m, and the linear mass density of the string is $\mu = 1.6$ g/cm. The oscillator has a frequency of $f = 120$ Hz. The amplitude of the motion at P is small enough to consider it a node. A node also exists at Q.



1. What mass m allows the oscillator to set up the fourth harmonic (standing wave) on the string?
2. What standing wave, if any, can be set for if we chose to hang a mass of $m = 1$kg instead?

2 *The "Orthogonality" of Sinusoids [Foundational]

In class, we stated that the normal mode functions $\sin(n\pi x/L)$ on the interval $[0, L]$ are **orthogonal**: their "inner product" vanishes unless the two modes are identical.

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \;=\; \begin{cases} 0 & n \neq m \\ \dfrac{L}{2} & n = m \end{cases} \tag{1}$$

We used this without proof to extract Fourier coefficients. Here you will derive it, understanding precisely why it works. You may find the following trig identities useful,

$$2\sin A \sin B = \cos(A-B) - \cos(A+B)\,, \qquad 2\cos A\cos B = \cos(A-B) + \cos(A+B)\,. \tag{2}$$

1. **The cross term: $n \neq m$.** Using the product-to-sum identity, show that for positive integers $n \neq m$:

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \;=\; 0. \tag{3}$$

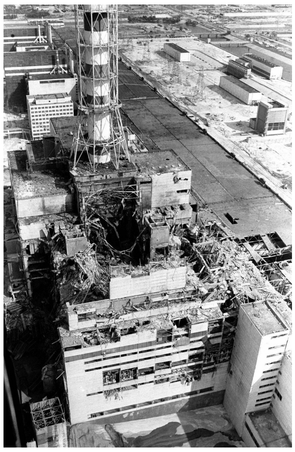
   Be explicit about why the boundary terms vanish: you should find that you are evaluating $\sin(n\pi)$ and $\sin(m\pi)$ for integer n, m, which are both zero. This is not a coincidence — it is precisely the fix-fix boundary conditions enforcing orthogonality.

2. **The diagonal term: $n = m$.** Show that

$$\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx \;=\; \frac{L}{2}. \tag{4}$$

   There are two clean ways to do this: (i) use the product-to-sum identity of Eq. (2) with $A = B$, or (ii) use the identity $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$ directly. Either is fine; the result is the same.

3 *The Chernobyl Disaster and The Heat Equation [Substantive]



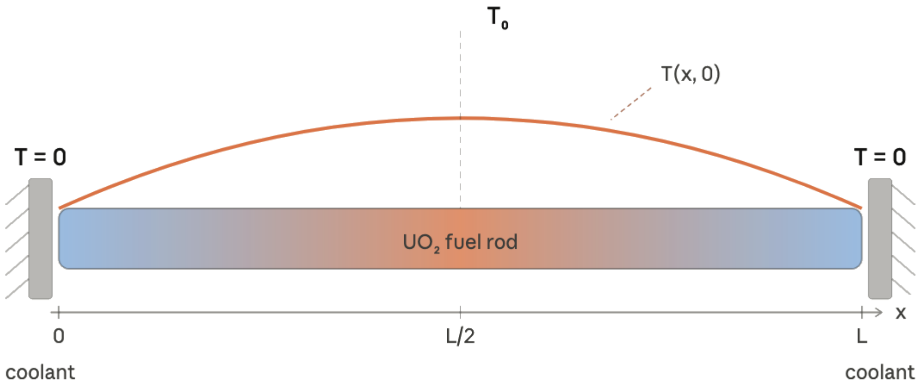
April 26, 1986. At 1:23 AM, operators at the Chernobyl Nuclear Power Plant located near Pripyat, Soviet Union (now Ukraine) pressed the emergency shutdown button, known as a *SCRAM.* Control rods were driven into the reactor core to absorb neutrons and halt the fission chain reaction. Due to a catastrophic design flaw, the rods briefly *increased* reactivity as they entered. Power surged to ∼30,000 times normal in three seconds. The reactor exploded. But the disaster did not end with the explosion. Even after fission stops, a reactor's fuel rods continues to generate heat. It drove the graphite moderator to catch fire, and lofting radioactive particles across Europe for ten days.

How does the temperature profile along a fuel rod evolve after SCRAM? Which part stays hottest longest, and when is it safe to approach? These are the questions you will answer — using mathematics you already know from the vibrating string.

Model a fuel rod (typically made of uranium dioxide UO_2) of length L as a 1D conductor. Both ends are held at $T = 0$ (excess above coolant temperature). The temperature $T(x,t)$ satisfies the **heat equation** (another kind of "wave equation"),

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}, \qquad T(0,t) = T(L,t) = 0, \tag{5}$$

where $\alpha = \kappa/(\rho c_p)$ is the thermal diffusivity.



A reactor running at steady state has a parabolic temperature profile — hottest at the centre where fission is most intense, cooled to zero at both ends by the coolant. At SCRAM ($t = 0$) we therefore take

$$T(x,0) \;=\; \frac{4T_0}{L^2}\,x(L-x), \tag{6}$$

where T_0 is the peak temperature at the midpoint.

a. Separating variables and identifying the eigenfunctions/normal modes.

Try a separable form solution $T(x,t) = X(x)G(t)$, similar to the idea of normal modes we had in class, in Eq. (5) and separate to obtain (dot is time derivative and prime is space derivative)

$$\frac{\dot G}{G} = \alpha\,\frac{X''}{X}\,.$$

(i) Since the LHS of this equation depends only on time, and RHS only on space, and they are equal for all (x,t), argue that both sides of the equation better be a constant. Call the constant $b = -\alpha\lambda$.

(ii) Argue that λ must be positive (consider what $\lambda \le 0$ would imply about the rod's temperature at large t).

(iii) Solve the spatial equation $X'' = -\lambda X$ with the boundary conditions $X(0) = X(L) = 0$ to find the allowed values λ_n and mode functions $X_n(x)$. You should recognize these immediately from class.

(iv) Solve the time equation to show that $G_n(t) = e^{-t/\tau_n}$ where

$$\tau_n = \frac{L^2}{\alpha n^2 \pi^2}.$$

This is exponential *decay*, not oscillation. In one sentence, identify the single structural difference between the heat equation and the wave equation that causes this.

(v) Write the general solution

$$T(x,t) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-t/\tau_n}. \tag{7}$$

How many free constants C_n does each mode carry, and why? Compare to the wave equation.

b. Fourier inversion: finding the coefficients. Use the Fourier inversion we have discussed in class and in Problem 2 of this tutorial to solve for the coefficients C_n for the initial temperature profile of Eq. (6),

$$C_n = \frac{2}{L}\int_0^L \frac{4T_0}{L^2}\, x(L-x)\, \sin\left(\frac{n\pi x}{L}\right) dx.$$

Evaluate this integral (integrate by parts twice, differentiating $x(L-x)$ each time) and show

$$C_n = \begin{cases} \dfrac{32\,T_0}{n^3\pi^3} & n \text{ odd} \\ 0 & n \text{ even.} \end{cases} \tag{8}$$

Explain why only odd n appear, using a symmetry argument: the parabola $x(L-x)$ is symmetric about $x = L/2$, while even-n modes are antisymmetric about $x = L/2$. Sketch one even and one odd mode shape to support your argument.

c. Numerical estimates for UO_2.

For a UO_2 fuel rod of $L = 4.0$ m, has $\alpha = 1.12 \times 10^{-6}\ \mathrm{m^2 s^{-1}}$ and a central temperature of $T_0 = 1200\,°\mathrm{C}$.

(i) Calculate the fundamental decay time τ_1. Express τ_1 in hours.

(ii) At time $t = \tau_1$, compute the ratio

$$\frac{C_3\, e^{-\tau_1/\tau_3}}{C_1\, e^{-\tau_1/\tau_1}}.$$

Since $\tau_3 = \tau_1/9$, you will find $e^{-9} \approx 10^{-4}$. What does this tell you about the temperature profile shape at late times?

(iii) In the single-mode approximation $T(x,t) \approx C_1 \sin(\pi x/L)\, e^{-t/\tau_1}$, find the time t_{safe} at which the mid-point temperature $T(L/2,\, t)$ falls below $100\,°\mathrm{C}$. Express in hours. Is t_{safe} an over- or underestimate of the true cooling time, given that our model ignores ongoing decay heat generation?

d. The deeper contrast: oscillation, decay, and quantum mechanics.

(i) The heat equation and the wave equation have identical spatial mode eigenfunctions $\sin(n\pi x/L)$ on the same domain. Yet one system oscillates forever and the other decays irreversibly. What property of the temporal factor $e^{-\lambda_n t}$ (heat) vs. $e^{\pm i\omega_n t}$ (wave) is responsible for the irreversibility of heat flow? (*Hint*: compare $|e^{-\lambda_n t}|$ and $|e^{\pm i\omega_n t}|$ as $t \to \infty$.)

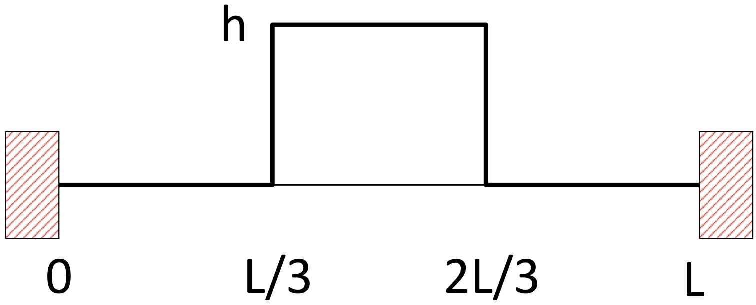
(ii) Remember we looked at the time-dependent Schrödinger equation for a quantum particle in a 1D box of length L in class?

$$i\hbar\,\frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \Psi}{\partial x^2}, \qquad \Psi(0,t) = \Psi(L,t) = 0.$$

Without solving it, use the analogy with Eq. (5) to write down the spatial eigenfunctions $\psi_n(x)$ and the form of the temporal factor $G_n(t)$ such that $\Psi(x,t) = \psi(x)G(t)$. (*Hint*: the spatial differential equation is identical; the time differential equation gains a factor of i.) Is quantum evolution reversible or irreversible? How can you tell from $|G_n(t)|$?

4 A Square Wave on a String [Foundational]

Consider a taut string of equilibrium length L, fastened at both ends. At time $t = 0$ the middle third is streched into a square pulse shape of height h as shown in the figure below. It is at rest before it is released at $t = 0$.



1. What is the shape of the string as a function of time? Express your result in terms of h, L, and the wave velocity v (and time, of course). You may assume no energy loss.
2. Does the string ever return to the initial shape? If so, when?

5 On Open Boundary Conditions [Basic]

In class we discussed normal modes of a string of length L and found it to be of the following form,

$$y(x,t) = (A\cos kx + B\sin kx)(C\cos\omega t + D\sin\omega t) \ , \tag{9}$$

where $\omega = vk$ and $v = \sqrt{T/\mu}$ is the speed of the wave on the string. Instead of looking at the "fix-fix" boundary conditions we looked at in class, we will now look at "fix-open" boundary conditions, where one end is clamped fixed and the other is free to move.

1. The left end is fixed and satisfies $y(0,t) = 0$. The right end is free to move. Think about it, the end point is a massless point and therefore the net force on it must vanish for an arbitrary, finite acceleration. Use these boundary conditions to find the allowed values of k on the string. *Hint:* Only discrete values will be allowed.
2. What frequencies are allowed on the string?
3. Now let's apply a similar idea to musical instruments where the air displacement acts as the oscillating variable. The bansuri is open at both ends. At an open end, the air displacement is free and the boundary condition is the same as that for an "open" string end. Show that for an open-open pipe, the allowed frequencies are $f_n = nv/2L$, the same as "fix-fix".
4. The *bansuri* (bamboo flute) is open at both ends; the *shehnai* is effectively closed at the reed end and open at the bell. Both are roughly the same playing length $L \approx 60$ cm. Using your result from (a) and the open-open result (which by symmetry gives the same allowed frequencies as fix-fix), write down the fundamental frequency of each instrument. What is the ratio $f_{\text{shehnai}}/f_{\text{bansuri}}$? What does this imply about the relative pitch of the two instruments for the same tube length?